

Week 1 Module

1 Syntan :-

In C language, Syntan is the set of rules that defines how to write valid C programs. It specifies how keywords, variables, operators, functions, and statements must be arranged.

Basic Syntan of a C Program:

```
#include <stdio.h>
```

```
int main() {
```

```
    printf("Hello World");
```

```
    return 0;
```

```
}
```

Important C Syntan Rules :-

- (i) Every statement ends with a Semicolon (;).
- (ii) C is case-sensitive (sum and Sum are different).
- (iii) Code blocks are enclosed in curly braces {}.
- (iv) Variables must be declared before they are used.
- (v) Comments can be written as:

// Single-line comment.

/* Multi-line comment */.

2. Conditions and Loops :-

2. Conditions and loops :-

(i) Conditional Statements: Conditional statements allow a program to make decisions based on a condition.

(a) if Statement : Executes a block of code only if the condition is true.

Syntax: `if (condition) {
 // statement
}`

(b) if-else statement : Executes one block if the condition is true, otherwise another block.

Syntax: if (condition) {

```

} else {
    // Statement
}

```

11 Statement

3.

(c) else if ladder: Checks multiple conditions

Syntax: if (condition) {

11 Statement

} else if (Condition 2) {

// Statement

```
} else {
```

// Statement

3.

(d) Switch Statement : Used to select one of many options.
Syntax: Switch (expression) {.

Case Value 1 :

// Statements

break ;

Case Value 2 :

// Statements -

break ;

default :

// Statements

};

(ii) Loops : Loops are used to repeat a block of code.

(a) for loop : Used when the number of iterations is known.

Syntax: for (initialization; condition; increment) {.

// Statement

};

(b) while loop : Executes as long as the condition is true.

Syntax: while (condition) {.

// Statement

};

(c) do-while loop : Executes the loop body at least once, then checks the condition.

Syntax: do {

// Statements

} while (condition);

3. Programming Problems :-

(i) Square :-

```
#include <stdio.h>
```

```
int main () {
```

```
    int n, i, j;
```

```
    printf("Enter the size of the square:");
```

```
    scanf("%d", &n);
```

```
    for (i=1; i<=n; i++) {
```

```
        for (j=1; j<=n; j++) {
```

```
            printf("#");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

Output :-

Enter the size of the square : 5

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```



main.c

Output



```
1  #include <stdio.h>
2
3  int main() {
4      int n, i, j;
5
6      printf("Enter the size of the square:
7          ");
8
9      scanf("%d", &n);
10
11     for (i = 1; i <= n; i++) {
12         for (j = 1; j <= n; j++) {
13             printf("* ");
14         }
15         printf("\n");
16     }
17     return 0;
18 }
```




main.c

Output



Enter the size of the square: 5

* * * * *

* * * * *

* * * * *

* * * * *

* * * * *

=== Code Execution Successful ===

(ii) Hollow Square :

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, i, j;
```

```
    printf("Enter the size of the Hollow Square : ");
```

```
    scanf("%d", &n);
```

```
    for (i = 1; i <= n; i++) {
```

```
        for (j = 1; j <= n; j++) {
```

```
            if (i == 1 || i == n || j == 1 || j == n)
```

```
                printf("# ");
```

```
            else
```

```
                printf(" ");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

Output :

Enter the size of the Hollow Square : 5

```
* * * * *
*           *
*         *
*       *
*     *
* * * * *
```



main.c

Output



```
1  #include <stdio.h>
2
3  int main() {
4      int n, i, j;
5
6      printf("Enter the size of the square:
7          ");
8
9      scanf("%d", &n);
10
11     for (i = 1; i <= n; i++) {
12         for (j = 1; j <= n; j++) {
13             if (i == 1 || i == n || j == 1
14                 || j == n)
15                 printf("* ");
16             else
17                 printf("  ");
18         }
19         printf("\n");
20     }
```




Programiz

C Online Compiler

Programiz PRO

main.c

Output



Enter the size of the square: 5

```
* * * * *
*       *
*       *
*       *
*       *
* * * * *
```

=== Code Execution Successful ===

(iii) Triangle :

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, i, j;
```

```
    printf("Enter the no. of rows: ");
```

```
    scanf("%d", &n);
```

```
    for (i = 1; i <= n; i++) {
```

```
        for (j = 1; j <= i; j++) {
```

```
            printf("* ");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

Output :-

Enter the no. of rows: 5

```
*
**
***
****
*****
```



main.c

Output



```
1  #include <stdio.h>
2
3  int main() {
4      int n, i, j;
5
6      printf("Enter the number of rows: ");
7      scanf("%d", &n);
8
9      for (i = 1; i <= n; i++) {
10         for (j = 1; j <= i; j++) {
11             printf("* ");
12         }
13         printf("\n");
14     }
15
16     return 0;
17 }
```




main.c

Output



Enter the number of rows: 5

*

* *

* * *

* * * *

* * * * *

=== Code Execution Successful ===

(iv) Inverted Triangle

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, i, j;
```

```
    printf("Enter the no. of rows: ");
```

```
    scanf("%d", &n);
```

```
    for (i = n; i >= 1; i--) {
```

```
        for (j = 1; j <= i; j++) {
```

```
            printf("# ");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

Output :

Enter the no. of rows: 5

* * * * *

* * * *

* * *

* *

*



main.c

Output



```
1  #include <stdio.h>
2
3  int main() {
4      int n, i, j;
5
6      printf("Enter the number of rows: ");
7      scanf("%d", &n);
8
9      for (i = n; i >= 1; i--) {
10         for (j = 1; j <= i; j++) {
11             printf("* ");
12         }
13         printf("\n");
14     }
15
16     return 0;
17 }
```




Programiz

C Online Compiler

Programiz PRO

main.c

Output



Enter the number of rows: 5

* * * * *

* * * *

* * *

* *

*

=== Code Execution Successful ===

(N) Pyramid.

```
#include <stdio.h>
```

```
int main () {
```

```
    int n, i, j;
```

```
    printf("Enter the no. of rows: ");
```

```
    scanf("%d", &n);
```

```
    for (i = 1; i <= n; i++) {
```

```
        for (j = 1; j <= n - i; j++) {
```

```
            printf(" ");
```

```
        }
```

```
        for (j = 1; j <= (2 * i - 1); j++) {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

Output :

Enter the no. of rows: 5

```

      *
     ***
    *****
   ********
  *********
 
```



main.c

Output



```
1  #include <stdio.h>
2
3  int main() {
4      int n, i, j;
5
6      printf("Enter the number of rows: ");
7      scanf("%d", &n);
8
9      for (i = 1; i <= n; i++) {
10
11          // Print spaces
12          for (j = 1; j <= n - i; j++) {
13              printf(" ");
14          }
15
16          // Print stars
17          for (j = 1; j <= (2 * i - 1); j++)
18              {
19                  printf("*");
20              }
21          printf("\n");
22      }
23
24      return 0;
25  }
```




Programiz

C Online Compiler

Programiz PRO

main.c

Output



Enter the number of rows: 5

*

=== Code Execution Successful ===

(vi) Inverted Pyramid

```
#include <stdio.h>
```

```
int main () {
```

```
    int n, i, j;
```

```
    printf("Enter the no. of rows: ");
```

```
    scanf("%d", &n);
```

```
    for(i=n; i>=1; i--) {
```

```
        for(j=1; j<=n-i; j++) {
```

```
            printf(" ");
```

```
        }
```

```
        for(j=1; j<=(2*i-1); j++) {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

Output :

Enter the no. of rows: 5

```
* * * * *
 * * * *
  * * * *
   * * *
    * *
     *
```



main.c

Output



```
1  #include <stdio.h>
2
3  int main() {
4      int n, i, j;
5
6      printf("Enter the number of rows: ");
7      scanf("%d", &n);
8
9      for (i = n; i >= 1; i--) {
10
11          // Print leading spaces
12          for (j = 1; j <= n - i; j++) {
13              printf(" ");
14          }
15
16          // Print stars
17          for (j = 1; j <= (2 * i - 1); j++)
18              {
19                  printf("*");
20              }
21          printf("\n");
22
23      }
24
25      return 0;
26 }
```



main.c

Output



```
Enter the number of rows: 5
```

```
*****
```

```
*****
```

```
*****
```

```
***
```

```
*
```

```
=== Code Execution Successful ===
```


(vii) Half Diamond

```
#include <stdio.h>
```

```
int main() {
```

```
    int n, i, j;
```

```
    printf("Enter the no. of rows: ");
```

```
    scanf("%d", &n);
```

```
    for (i=1; i<=n; i++) {
```

```
        for (j=1; j<=i; j++) {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    for (i=n-1; i>=1; i--) {
```

```
        for (j=1; j<=i; j++) {
```

```
            printf("*");
```

```
        }
```

```
        printf("\n");
```

```
    }
```

```
    return 0;
```

```
}
```

Output :

Enter the no. of rows: 5.

```
*
**
***
****
*****
****
***
**
*
*
```



main.c

Output



```
1  #include <stdio.h>
2
3  int main() {
4      int n, i, j;
5
6      printf("Enter the number of rows: ");
7      scanf("%d", &n);
8
9      // Upper half
10     for (i = 1; i <= n; i++) {
11         for (j = 1; j <= i; j++) {
12             printf("* ");
13         }
14         printf("\n");
15     }
16
17     // Lower half
18     for (i = n - 1; i >= 1; i--) {
19         for (j = 1; j <= i; j++) {
20             printf("* ");
21         }
22         printf("\n");
23     }
24
25     return 0;
26 }
```



main.c

Output



Enter the number of rows: 5

*

* *

* * *

* * * *

* * * * *

* * * *

* * *

* *

*

=== Code Execution Successful ===

(viii) Diamond.

```
#include <stdio.h>
```

```
int main () {
```

```
    int n, i, j;
```

```
    printf("Enter the no. of rows: ");
```

```
    scanf("%d", &n);
```

```
    for (i=1; i<=n; i++) {
```

```
        for (j=1; j<=n-i; j++)
```

```
            printf(" ");
```

```
        for (j=1; j<=(2*i-1); j++)
```

```
            printf("*");
```

```
        printf("\n");
```

```
    }
```

```
    for (i=n-1; i>=1; i--) {
```

```
        for (j=1; j<=n-i; j++)
```

```
            printf(" ");
```

```
        for (j=1; j<=(2*i-1); j++)
```

```
            printf("*");
```

```
        printf("\n");
```

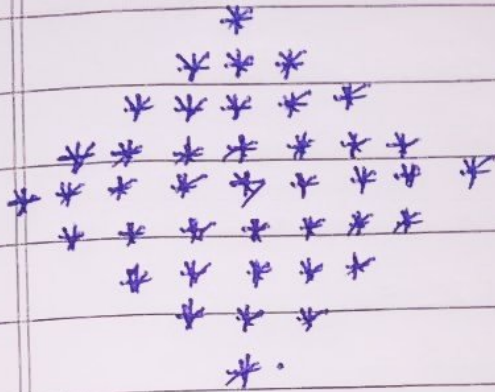
```
    }
```

```
    return 0;
```

```
}
```


Output:

Enter the no. of rows: 5.





main.c

Output



```
1  #include <stdio.h>
2  int main() {
3      int n, i, j;
4      printf("Enter the number of rows: ");
5      scanf("%d", &n);
6      // Upper half
7      for (i = 1; i <= n; i++) {
8          // Print spaces
9          for (j = 1; j <= n - i; j++)
10             printf(" ");
11         // Print stars
12         for (j = 1; j <= (2 * i - 1); j++)
13             printf("*");
14
15         printf("\n");
16     }
17     // Lower half
18     for (i = n - 1; i >= 1; i--) {
19         // Print spaces
20         for (j = 1; j <= n - i; j++)
21             printf(" ");
22
23         // Print stars
24         for (j = 1; j <= (2 * i - 1); j++)
25             printf("*");
26         printf("\n");
27     }
28     return 0;
29 }
```

Run



main.c

Output



Enter the number of rows: 5

*

*

=== Code Execution Successful ===